HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Owned module performance metrics across safety, quality, and output for a high-volume semiconductor manufacturing line, reducing process variation by **15%** through statistical analysis.
- Sustained and optimized wafer bumping and interconnect formation processes, ensuring stable, repeatable, high-quality output for critical product ramps.
- Partnered with 24x7 technician teams to rapidly contain and resolve equipment and process issues, minimizing production downtime.
- Led structured problem-solving initiatives across cross-functional teams, including process engineers and equipment specialists, to address root causes of yield deviations.
- Collaborated with global Virtual Factory partners to align manufacturing standards and deploy best practices across multiple fab sites.
- Supported technology ramps and tool installations for new semiconductor packaging processes, ensuring smooth transition into high-volume manufacturing.
- Containerized core Python data processing services for fab analytics using **Docker** and deployed them to an **AWS EKS** cluster, improving deployment consistency.

SKILLS

Domain Knowledge – Semiconductor manufacturing, Wafer packaging, Bumping, Interconnect formation, Process control, Process improvement, Process characterization, Manufacturing operations

Technical Skills – Data analysis, Statistical methods, Structured problem solving, Root cause analysis, Continuous improvement, Python, SQL, JMP

Tools & Technologies – AWS, Docker, Jenkins, Git, Linux, Splunk, Dynatrace, Jira, Confluence

Programming Languages – Python, Java, SQL

Databases – PostgreSQL, MySQL, Oracle

PROJECTS

Semiconductor Process Optimization Initiative (HCL Technologies)

- Applied statistical methods and **JMP** to analyze historical production data for a critical wafer packaging module, identifying key factors contributing to yield loss.
- Developed Python scripts to automate data extraction and preliminary analysis from fab operational logs, reducing manual analysis time by **40%** for the engineering team.
- Presented findings and recommended process adjustments to cross-functional teams, contributing to a sustained **12%** improvement in module output consistency.

Fab Data Pipeline and Monitoring (HCL Technologies)

- Engineered a Python-based data pipeline on **AWS Lambda** to ingest and process real-time sensor data from manufacturing equipment, enabling faster issue detection.
- Implemented monitoring dashboards using **Splunk** to visualize key performance indicators and alert on anomalies, ensuring timely intervention for critical process deviations.
- Collaborated with operations stakeholders to define data requirements and refine alerting thresholds, improving the accuracy of fault detection.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)